Exercise 2

1)
```
SELECT DISTINCT department
FROM            Table_students ;
```

Table students

department
IT
HR
Finance

2)
```
SELECT ~~Department~~ department,
           AVG (age) AS avg_age
FROM       ~~Table~~_students
GROUP BY   department
```

Table students

department	avg_age
IT	20.5
HR	22
Finance	23

3)
```
SELECT department,
       COUNT (*) AS Student_count
FROM   students
GROUP BY  department
HAVING COUNT(*) > 1 ;
```

Students

department	student count
IT	2
HR	2

4)
```
SELECT student_id,
       name,
       age,
       department
FROM   Students
WHERE age BETWEEN 21 AND 23;
```

Students

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5)
```
SELECT student_id,
       name,
       age,
       department
FROM   students
WHERE  department IN ('IT', 'HR')
 AND age > 21;
```

Students

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6)
```
SELECT   department,
         SUM(credits) AS total_credits
FROM     Courses
GROUP BY department
HAVING   SUM(credits) > 5;
```

Courses

department	total_credits
IT	11

7)
```
SELECT  course_id,
        course_name,
        department, credits
FROM    Courses
WHERE   credits != 4 ;
```

Courses

Course id	Course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8)
```
SELECT   course_id,
         Course_name,
         credits
FROM     Courses
ORDER BY credits DESC
LIMIT    3 ;
```

Courses

Course_id	Course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9)
```
SELECT  MAX (grade) AS max_grade,
        MIN (grade) AS min_grade,
        AVG (grade) AS avg_grade
FROM    enrollments;
```

enrollments

max_grade	min_grade	avg_grade
90	78	84.6

10)
```
SELECT  course_id,
        COUNT(*) AS enrollment_count
FROM    enrollments
GROUP BY  course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11) SELECT department,
SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus
FROM Salaries
GROUP BY department;

Salaries

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 SELECT department,
AVG (salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG (salary) > 55000 ;

Salaries

department	avg-salary
IT	61000
Finance	70000

3) SELECT employee-id,
name ,
salary ,
bonus AS total_compensation
FROM Salaries
WHERE (salary and bonus) > 60 000 ;

salaries

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

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```
SELECT   department,
         SUM(budget) AS total_budget,
         AVG(budget) AS avg_budget
FROM     projects
GROUP BY   department
HAVING     AVG(budget) >70000;
```

projects

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

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```
SELECT   project_id,
         project_name,
         department,
         budget
FROM     projects
WHERE    budget BETWEEN 50000 AND 120000
AND department != 'Marketing';
```

projects

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000